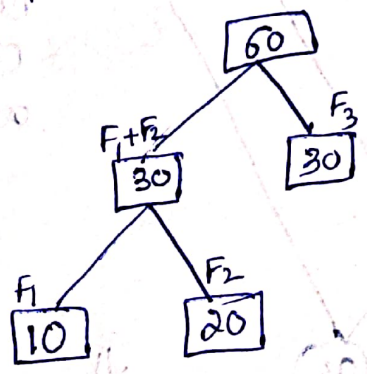
Optimal Merge Pattern

Algorithm

1. Create Min heap with 'n' files
2. At each level remove two minimum 'a' and 'b' from min heap and put a + b again into heap

F_1	F_2	F_3
10	20	30

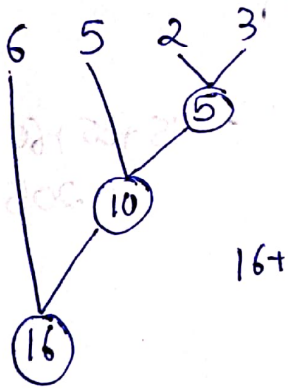


Step 1) use min first

Step 2) Update every time

Que-1

List	A	B	C	D
Size	6	5	2	3



$16+10+5 = \underline{31}$

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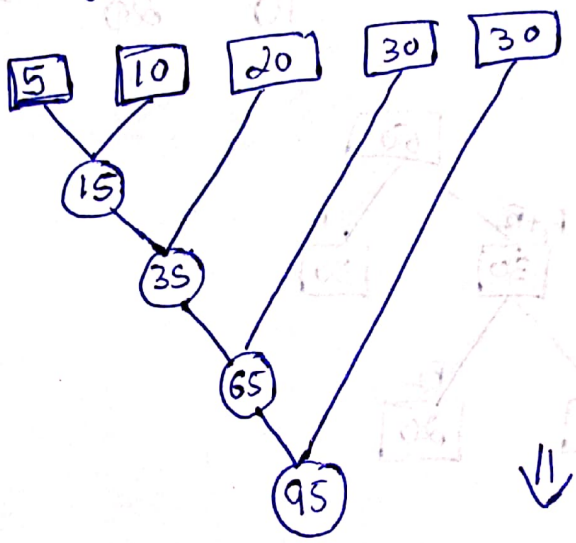
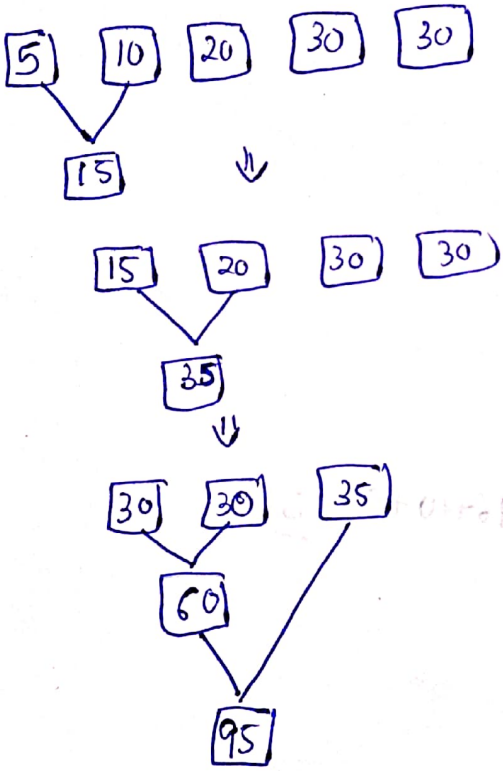
Qu-2

list →	x_1	x_2	x_3	x_4	x_5
Size →	20	30	10	5	30

Step-1) Arrange in increasing (Asc.) order



No this is wrong
Check everything result
among and then add



$15 + 35 + 60 + 95$
$= 205$

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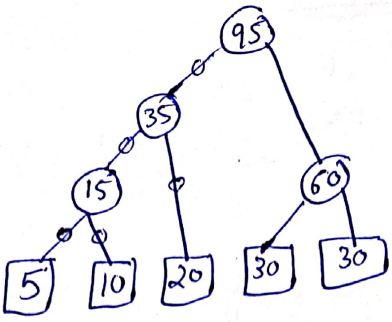

$\boxed{5} \times 3 + \boxed{10} \times 3 + \boxed{20} \times 2 + \boxed{30} \times 2 + \boxed{30} \times 2$

$15 + 30 + 40 + 60 + 60$

$= \boxed{205}$

$\Sigma\, d_i x_i$

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